

Operational Coherence Maintenance and the Quantum–Classical Boundary: Formal Definitions, Falsifiable Protocols, and an Outlook for Cognitive Systems

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Scope note. No collapse mechanism is proposed; the Born rule is not derived; no claims are made about phenomenological consciousness. **Series note (v2).** This is the integrative, protocol-facing paper of the coherence-maintenance series (ai.viXra:2512.0060/0061/0064/0070/0071/0072/0073, all at v2, 2026, with verification suites in a single companion repository). v1 of this paper stated its central inequality as proved here via a paired-strategy infimum; that formulation was vacuous, and the corrected operational hierarchy is imported from 2512.0061 (v2) (Theorem 3.1); what *is* proved here (Appendix A) is the unconditional entropy-production bound.

Abstract

Maintaining quantum coherence against uncontrolled open-system dynamics is a control task with unavoidable thermodynamic cost. In a finite-dimensional setting with battery-assisted thermal operations at bath temperature T , we present the corrected operational hierarchy for maintenance power: an unconditional lower bound $P_{\min}(\rho) \geq k_B T \sigma(\rho)$ with $\sigma(\rho)$ the entropy production rate of the target (proved in Appendix A); the coherence-power bound $P_{\min} \geq k_B T \dot{C}_{\text{loss}}(\rho)$ under diagonal contraction; and incremental (extra) power bounds relative to thermodynamically efficient population-maintenance baselines, which become assumption-free exactly when the pinched target is stationary — e.g. under pure dephasing, the very dissipator used in this paper’s numerical protocol. Here $C(\rho) = S(\rho \| \Delta[\rho])$ is relative-entropy coherence to energy pinching and $\dot{C}_{\text{loss}}(\rho) := -\frac{d}{dt}C(\rho_t)|_{t=0}$. These statements are operational, observer-independent, and geometry-free. We then present the Rate Inheritance Principle (RIP) as the falsifiable dynamical bridge between static clustering and decoherence rates, with its July-2026 status: weak form a lemma under explicit hypotheses; strong form derived, with a frequency-resolved squared-amplitude exponent, in an exactly solvable quasi-free local-sink class; failure through near-zero Bohr-frequency channels realized within the secular Davies class. The exact $\omega = 0$ Dirichlet identity underlying the failure mode is quoted in its verified form. We provide falsifiable protocols distinguishing one-shot work from sustained maintenance power, including a numerical stress test (distance-independent influence floor without an interface vs. collar-induced suppression) in a transverse-field Ising chain with remote dephasing, whose fixed-horizon metric and unitary-baseline subtraction already satisfy the proxy-validation discipline the series later codified. Finally, an explicitly speculative Outlook connects the resource boundary to the Free-Energy Principle for resource-limited agents, at a methodological (non-phenomenological) level.

1 Introduction

Decoherence explains the passive suppression of interference in open quantum systems [1, 2, 3]. This paper focuses on a complementary, operational question: what minimal control resources

are required to *prevent* coherence loss and keep a desired state stationary against uncontrolled dynamics?

We work within an explicit thermodynamic control model (battery-assisted thermal operations) and present the corrected hierarchy of lower bounds on maintenance power, proved in [19] (v2) and, for the unconditional part, in Appendix A. Separately, gapped systems exhibit static clustering [9, 10, 11]; whether dynamical decoherence rates inherit that suppression is the falsifiable Rate Inheritance Principle, whose status has moved substantially since v1 (Section 4). Together these motivate an operational reading of the quantum–classical boundary as a *resource boundary* in state space: coherence is “effectively classical” for an agent when maintaining it exceeds the agent’s power budget [20].

Non-claims (read first). No collapse mechanism; no Born-rule derivation; no claims about phenomenological consciousness. Any “cut” discussed here is operational (a feasibility boundary), not ontological.

Remark 1.1 (What changed in v2). v1 defined the incremental power as an infimum over *unlinked* strategy pairs and claimed the inequality $P_{\text{extra}} \geq k_B T \dot{C}_{\text{loss}}$ as proved here. That definition is vacuous — the infimum is $-\infty$, since population-maintenance strategies can burn battery at any rate — and v1’s Appendix A obtained the bound by subtracting two *lower* bounds (its Step 4), which is invalid; the same error pattern affected the v1 of the companion work theorem and is documented there [19]. The corrected statements are Theorem 3.1 below. What v1’s appendix does prove — kept here as Appendix A — is the *unconditional* entropy-production bound, which needs no pairing at all.

2 Operational framework

2.1 States, free energy, coherence

X finite-dimensional, $\gamma_X := e^{-\beta H_X} / \text{Tr } e^{-\beta H_X}$, $\beta = 1/(k_B T)$; $S(\rho \parallel \sigma) := \text{Tr}[\rho(\log \rho - \log \sigma)]$ (support convention); $F_X(\rho) := k_B T S(\rho \parallel \gamma_X)$; energy pinching $\Delta[\rho] := \sum_n \Pi_n \rho \Pi_n$; coherence $C(\rho) := S(\rho \parallel \Delta[\rho])$ (nats); loss rate $\dot{C}_{\text{loss}}(\rho) := -\frac{d}{dt} C(e^{t\mathcal{L}} \rho) |_{t=0}$; entropy production $\sigma(\rho) := -\frac{d}{dt} S(e^{t\mathcal{L}} \rho \parallel \gamma_S) |_{t=0} \geq 0$ (Spohn [5]), with the pinching split $\sigma = \dot{F}_{\text{diag}}^\Delta + \dot{C}_{\text{loss}}$.

2.2 Control model

Strategies are stroboscopic: free evolution $e^{\Delta t \mathcal{L}}$ alternating with battery-assisted thermal operations — global unitaries U with $[U, H_S + H_B + H_W] = 0$ acting on $\rho_S \otimes \gamma_B \otimes \sigma_W$ — with work accounted as battery free-energy *consumption* (sign convention of [19] v2). $P(s)$ is the asymptotic consumption rate; $\text{Ctrl}(\rho)$ the maintaining strategies.

2.3 Incremental power, corrected definitions

Definition 2.1 (ε -efficient baseline [19]). $s_{\text{diag}} \in \text{Ctrl}(\Delta[\rho])$ is ε -efficient if $P(s_{\text{diag}}) \leq k_B T \sigma(\Delta[\rho]) + \varepsilon$, i.e. it exceeds its own task’s thermodynamic floor by at most ε .

Definition 2.2 (Extra power). $P_{\text{extra}}(\rho) := P_{\min}(\rho) - P_{\min}(\Delta[\rho])$ (difference of infima, both over Ctrl sets assumed non-empty, the second finite).

Remark 2.3 (Why not unlinked pairs). $\inf_{(s, s_{\text{diag}})} [P(s) - P(s_{\text{diag}})] = -\infty$: fix s and let the baseline append battery-burning cycles (unitaries on WB dumping battery free energy into the bath act trivially on S and preserve maintenance of $\Delta[\rho]$). v1’s definition and its “baseline matching” remark did not exclude this; efficiency of the baseline is the operative constraint, and it is precisely what a careful experimental comparison implements (Appendix B).

3 Main bounds: the corrected hierarchy

Theorem 3.1 (Maintenance-power hierarchy). *Under (A1)–(A4) and (A6) of v1 (finite dimensions; GKLS with thermal fixed point; energy-conserving strokes with thermal bath inputs; battery free-energy accounting; full-rank ρ):*

- (a) (Unconditional; proved in Appendix A) $P_{\min}(\rho) \geq k_B T \sigma(\rho)$.
- (b) (Coherence power; [19] v2) *under diagonal contraction (automatic for Davies generators), $P_{\min}(\rho) \geq k_B T \dot{C}_{\text{loss}}(\rho)$.*
- (c) (Extra power; [19] v2) *against ε -efficient baselines, under population autonomy, $P(s) - P(s_{\text{diag}}) \geq k_B T \dot{C}_{\text{loss}}(\rho) - \varepsilon$ for every $s \in \text{Ctrl}(\rho)$; and in the free-baseline case — $\mathcal{L}(\Delta[\rho]) = 0$ and $\Delta[\mathcal{L}(\rho)] = 0$, e.g. pure dephasing — $P_{\text{extra}}(\rho) = P_{\min}(\rho) \geq k_B T \dot{C}_{\text{loss}}(\rho)$ with no further assumptions.*

Remark 3.2 (The paper’s own protocols live in the free-baseline case). Protocol B below (and the numerical stress test) uses a *dephasing* dissipator. For dephasing-dominated platforms the pinched target is stationary, the population baseline is free, and the headline inequality $P_{\text{extra}} \geq k_B T \dot{C}_{\text{loss}}$ is a theorem with no baseline or contraction assumptions — v1’s claim was thus *true in its own experimental setting* for the corrected reason.

Interpretation is unchanged from v1: the bounds are operational, observer-independent, geometry-free; C is in nats, so $k_B T \dot{C}_{\text{loss}}$ is a power. v1’s regularity assumption (A5) is not needed for Theorem 3.1 and is dropped.

4 Static clustering and the dynamical bridge (RIP)

Gapped systems cluster statically [9, 10]; massive-field correlators carry Bessel envelopes $K_\nu(m\epsilon)$ [18]. Whether *rates* inherit these envelopes is the Rate Inheritance Principle, formalized in [23] (v2): the weak (upper-envelope) form is a lemma under explicit integrability hypotheses; the strong (envelope-class) form is a falsifiable conjecture. Status as of this version: *derived* — with the frequency-resolved, squared-amplitude exponent $\kappa(\epsilon) \propto e^{-2q(\omega_b)\epsilon}$, $\cosh q(\omega) = (\mu^2 + 4 - \omega^2)/(4\mu)$ — in an exactly solvable quasi-free local-sink class, verified against exact Liouvillian rapidities [20]; *failure realized* through near-zero-frequency channels within the secular Davies model class, whose delocalized jump operators sustain a persistent projected rate floor [21].

4.1 A concrete failure mode: near-zero-frequency floors

For a Davies generator with coupling S and an active $\omega = 0$ channel, the exact identity

$$\mathcal{E}_\sigma^{(0)}(O) = \frac{\gamma(0)}{2} \|[S(0), O]\|_{2,\sigma}^2 \quad (1)$$

(KMS norm; proved and machine-verified in [24] v2, where the general- ω decomposition requires modular weights and is *not* of commutator form) shows that whenever $S(0)$ retains non-decaying overlap with observables at distance ϵ — e.g. extensive conserved components, or the generic secular delocalization measured in [21] — a distance-independent rate floor arises despite static clustering. Floors are model-dependent (a conditional design principle, as in v1), and particularly relevant with persistent low-frequency channels.

5 Falsifiable protocols

Protocol A (one-shot vs sustained). Distinguish the one-shot work of shaping a state from the sustained power of holding it: the former scales with free-energy differences, the latter with $k_B T \dot{C}_{\text{loss}}$ (Theorem 3.1); geometric interfaces enter the two differently (amplitude² vs rate; see [19] §8 and [20]).

Protocol B (interface geometry: floor vs collar). As in v1, with the local X -basis pinching proxy C_x and the *fixed-horizon* influence metric

$$D_T(L) := \max_{t \in [0, T]} |\overline{C_x^{\text{noisy}}}(t; L) - C_x^{\text{unitary}}(t; L)|, \quad (2)$$

comparing uniform-link and collar-link geometries at matched microscopic parameters (TFIM, $N_{\text{tot}} = 14$, $h/J = 1.5$, $\gamma = 0.2$, dephasing jump at $j_{\text{diss}} = L + 1$, $T = 3.0$): uniform links show an influence floor (no suppression with L); collar links ($J_{\text{buf}} \ll J$) show clear suppression, with fitted rates μ increasing as the collar opacifies (v1 Table 1 retained; parameters fully declared).

Protocol B is an influence-suppression *diagnostic*, not by itself a direct measurement of the energy-pinching coherence appearing in Theorem 3.1: its readout uses the local X -basis pinching proxy C_x , and connecting the two requires calibrating the chosen pointer pinching to the relevant energy/pointer sector. Its role is to test the geometry-to-rate interface; Protocol A tests the thermodynamic maintenance inequality directly.

Remark 5.1 (Proxy discipline (v2)). The series’ methodology [20, 21, 23] requires rate proxies to be validated: fixed absolute horizons (not co-moving windows), explicit baseline subtraction, and gap-removed/geometry controls. Protocol B already complies with the first two by construction — D_T is fixed-horizon and subtracts the unitary baseline — and its uniform-vs-collar comparison plays the role of the geometry control; a critical-point sweep ($h \rightarrow J$) is the recommended third control for any platform implementation. This is worth stating because the v1 evidence of two companion papers failed precisely these checks and was withdrawn [20, 21].

6 The quantum–classical boundary as a resource boundary

Combining Theorem 3.1 with rate behavior: where rate inheritance holds (local-sink, sub-gap regimes), isolation is exponentially cheap in the interface width and the cost of coherence lifetime is logarithmic [20]; where floors persist ($\omega \simeq 0$ channels, within the Davies class), no separation reduces the bill and a finite budget enforces effectively classical descriptions — an operational cut [20, 22]. Below budget, no admissible strategy can maintain, exploit, or reverse the coherence over the relevant operational timescale.

7 Outlook for cognitive systems (speculative)

Unchanged in substance from v1, with the corrected anchoring. Cognitive systems are resource-limited physical agents; the Free-Energy Principle [14, 15] concerns what representations agents should maintain, while Theorem 3.1 bounds what maintaining them *costs* under explicit control primitives. The conditional multi-objective reading

$$\min_{q, \pi} \mathbb{E}_\pi [F_{\text{var}}(q)] + \lambda P_{\text{extra}}(q, \pi), \quad \lambda > 0, \quad (3)$$

with P_{extra} now in its corrected sense (efficient-baseline or free-baseline), predicts: fragile encodings that lower F_{var} either raise measured maintenance power or are abandoned once the budget is capped — a falsifiable trade-off in synthetic agents with explicit energy/compute budgets, methodologically continuous with metabolic-cost results in neuroscience [16, 17]. No claims about phenomenological consciousness, nor about quantum coherence in brains.

8 Limitations

Theorem 3.1 assumes finite dimensions, GKLS dynamics with thermal fixed point, bath thermal inputs, and stroboscopic battery-assisted thermal operations; extensions to strong coupling, non-Markovian regimes, or algebraic (Type III) settings are conditional interfaces [19, 18]. RIP remains open exactly for interacting gapped buffers with local dissipation [20]. The cognitive Outlook is methodological only.

A Proof of Theorem 3.1(a)

This is the valid core of v1’s Appendix A (its Steps 0–3 and 5; the paired Step 4 is removed, cf. Remark 1.1).

Step 0–1. With $\Gamma := \gamma_S \otimes \gamma_B \otimes \gamma_W$ and $[U, H_S + H_B + H_W] = 0$, $UTU^\dagger = \Gamma$, so $S(U\rho U^\dagger \| \Gamma) = S(\rho \| \Gamma)$ per stroke.

Step 2. Product-reference decomposition $S(\rho_{SBW} \| \Gamma) = \sum_X S(\rho_X \| \gamma_X) + I_\Gamma(\rho_{SBW})$ with $I_\Gamma \geq 0$; applied to input (product) and output states, this gives the per-stroke work bound $W \geq F_S(\rho'_S) - F_S(\rho_S)$, where W is battery free-energy consumption.

Step 3. Pinching Pythagoras: $S(\rho \| \gamma_S) = S(\Delta[\rho] \| \gamma_S) + C(\rho)$, i.e. $F_S = F_S \circ \Delta + k_B T C$.

Step 5. For any $s \in \text{Ctrl}(\rho)$ with cycle time Δt : the correction must implement $e^{\Delta t \mathcal{L}}(\rho) \mapsto \rho$, so by Step 2 its per-cycle consumption obeys $W(\Delta t) \geq F_S(\rho) - F_S(e^{\Delta t \mathcal{L}} \rho) = k_B T [S(\rho \| \gamma_S) - S(\rho_{\Delta t} \| \gamma_S)]$. Dividing by Δt , taking limsup over cycles and then $\Delta t \rightarrow 0$: $P_{\min}(\rho) \geq k_B T \sigma(\rho) \geq 0$, nonnegativity by Spohn [5]. With Step 3, $\sigma = \dot{F}_{\text{diag}}^\Delta + \dot{C}_{\text{loss}}$, from which (b) follows under diagonal contraction; (c) is proved in [19] (v2). $\square$

B Experimental calibration and power proxies

As in v1, with one clarification. Wall-plug comparisons of two strategies “implemented with identical primitives except for the coherence-stabilizing component” estimate $P_{\text{wall}}(s) - P_{\text{wall}}(s_{\text{diag}})$; by Theorem 3.1(c) this lower-bounds $k_B T \dot{C}_{\text{loss}} - \varepsilon$ *provided the baseline is thermodynamically efficient* — identical primitives is exactly the experimental implementation of ε -efficiency, and in dephasing-dominated platforms the baseline is free and the proviso is automatic (Remark 3.2). Thermal anchoring and error budgets as in v1.

C Changes relative to version 1

1. v1’s Theorem 1 and its Appendix A proof are corrected: the unlinked-pairs P_{extra} was vacuous and Step 4 subtracted lower bounds (Remark 1.1, Remark 2.3); v2 states the corrected hierarchy (Theorem 3.1), proves the unconditional part here (Appendix A = v1’s valid Steps 0–3, 5), and imports (b), (c) from [19] (v2). v1’s assumption (A5) is dropped as unnecessary.
2. Remark 3.2: the paper’s own protocols (dephasing) sit in the free-baseline case, where the headline inequality is assumption-free.
3. RIP section updated to the July-2026 status (derived quasi-free with squared exponent [20]; Davies-class floors [21]; lemma-level weak form [23]); the $\omega = 0$ identity is quoted in its verified form with the caveat that the general- ω decomposition is *not* of commutator form [24].

4. Proxy-discipline remark added to Protocol B (Remark 5.1); its fixed-horizon metric and unitary-baseline subtraction already comply, and the critical-point control is recommended for platform implementations. v1's Figure 1/Table 1 results are retained with fully declared parameters.
5. Series references added (v1 cited only external literature); calibration appendix tied to ε -efficient baselines; work sign fixed to the consumption convention.

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